

ISHAN SHAH

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EDUCATION

University of Waterloo

Bachelor of Computer Science

EXPERIENCE

Axibo

May 2026 – Present

Research Engineer

Cambridge, ON

- Built a robot learning stack for **bimanual manipulation and humanoid robots**, integrating **UMI-style collection**, synchronized multi-camera demonstration collection, LeRobot dataset generation, evaluation pipelines, and VLA deployment.
- Deployed OpenPI $\pi_{0.5}$ VLA models on real-world **PiPER-X and humanoid platforms**; improved robustness through relative action representations, temporal observation histories, and dynamic action chunks.
- Combined **WARP-BC** demonstration filtering with **DSRL** policy refinement to build a more data-efficient VLA training pipeline, lifting manipulation success rate **8%** and inference throughput **17%**.
- Implemented $\pi_{0.6}$ **RECAP** value-guided learning into the training loop to improve trajectory optimization and data utilization, raising task success rate by **10%**.
- Closed the real-time control loop with latency-aware action alignment and temporal interpolation, taking effective policy throughput to **$1.7\times$** for more reliable deployment.

WAT.ai

Jan. 2026 – July 2026

Robotics Researcher

Waterloo, ON

- Built simulation environments for training and evaluating sequential decision-making agents across a range of manipulation scenarios.
- Explored **VLM-based preference learning** to align robot policies with desired behavior under sparse rewards, improving task performance **15%** over the reward-shaped baseline.
- Built PyTorch RL pipelines (Actor-Critic) with human-in-the-loop data collection for iterative policy improvement.

IPMD

July 2025 – Sept. 2025

Software Engineer

San Mateo, CA

- Replaced custom auth logic with Flask-JWT-Extended, centralizing token management across 10 endpoints and fixing registration edge cases.
- Built a real-time computer-vision emotion analysis pipeline into a React (TypeScript) chat interface, surfacing live sentiment feedback.

TECHNICAL SKILLS

Languages: Python, C++

Frameworks: PyTorch, JAX, ROS2, OpenCV, NumPy

Simulation: MuJoCo, Isaac Lab, Genesis

Tools/Deployment: Linux, Git, Docker

PROJECTS

Human-in-the-Loop Teleoperation for Imitation Learning ↗ | *MuJoCo, MediaPipe, PyTorch*

- Designed a low-latency teleoperation system to control a simulated ALOHA bimanual robot using monocular webcam hand tracking, removing the need for specialized hardware.
- Built a real-time human-to-robot control pipeline using Inverse Kinematics, refining quaternion-based mappings to avoid gimbal lock.
- Developped an HDF5-based demonstration logging pipeline structured for training **Action-Conditioned Transformers (ACT) and Diffusion Policy models**.